Emerging Technologies for the Circular Economy

Lecture 10: Sustainability of Emerging Technologies

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TU Clausthal

Institute for Software and Systems Engineering

Two main perspectives

For any new technology *T*

Sustainability **of** *T*

VS

Usefulness of *T* **for** Sustainability

Introduction

CE/CS and Technology

	Holistic	Segmented
Optimist	**Reformist Circular Society** • *Assumptions*: reformed form of capitalism is compatible with sustainability and socio-techincal innovations can enable eco-economic decoupling to prevent ecological collapse. • *Goal:* economic prosperity and human well-being within the biophysical boundaries of the earth. • *Means*: technological breakthroughs, social innovations and new business models that improve ecological health, resource security, and material prosperity for all.	**Techcentric Circular Economy** • *Assumptions:* capitalism is compatible with sustainability and technological innovation can enable eco-economic decoupling to prevent ecological collapse. • *Goal:* sustainable human progress and prosperity without negative environmental externalities. • *Means:* economic innovations, new business models and unprecedented breakthroughs in CE technologies for the closing of resource loops with economic value creation.
Sceptical	**Transformational Circular Society** • *Assumptions:* capitalism is incompatible with sustainability and socio-technical innovation cannot bring absolute eco-economic decoupling to prevent ecological collapse. • *Goal:* a world of conviviality and frugal abundance for all, while fairly distributing the biophysical resources of the earth. • *Means:* complete reconfiguration of the current socio-political system and a shift away from productivist and anthropocentric worldviews to drastically reduce humanity's ecological footprint and ensure that everyone can live meaningfully, and in harmony with the earth.	**Fortress Circular Economy** • *Assumptions:* there is not alternative to capitalism and socio-technical innovation cannot bring absolute eco-economic decoupling to prevent ecological collapse. • *Goal:* maintain geostrategic resource security and earth system stability in global conditions where widespread resource scarcity and human overpopulation cannot provide for all. • *Means:* innovative technologies and business models combined with rationalized resource use, imposed frugality and strict migration and population controls.

Image recreated from: M. C. Friant, R. Salomone, W. J. V. Vermeulen (2020) – A Typology of Circular Economy Discourses: Navigating the Diverse Visions of a Contested Paradigm – Link

Poll:
Which of these is the way forward?

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Emissions Trading : The capitalist's solution

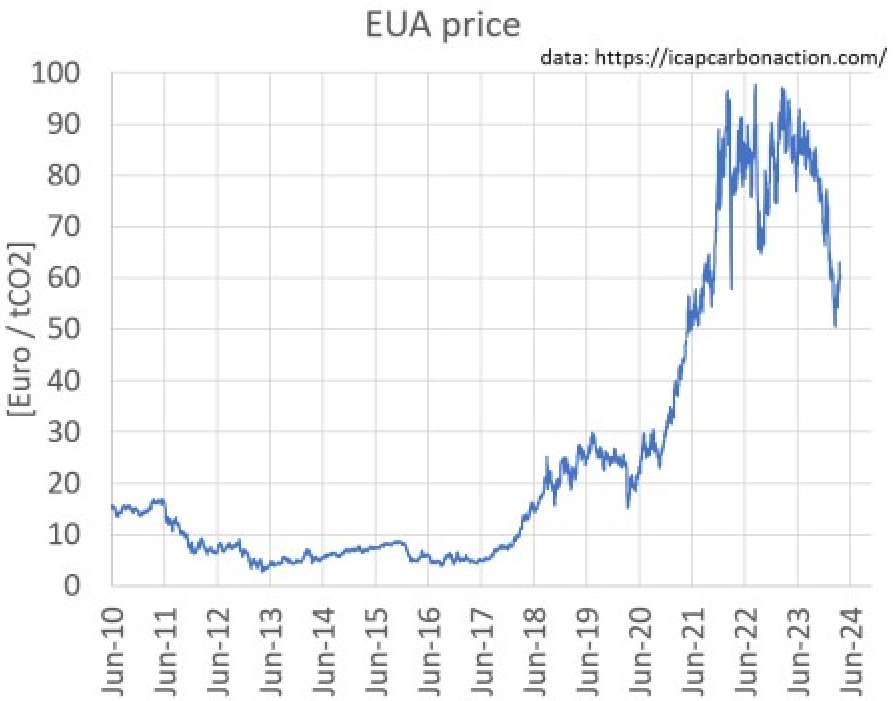
Quick Introduction

- An approach to limit climate change by creating a market with limited allowances for emissions. (Cap and Trade)
- Regulators set a quantitative total limit on the emissions produced by participating polluters (e.g., oil companies).
- A polluter having more emissions than their assigned quota MUST purchase the right to emit more.
- A polluter emitting fewer emissions than their assigned quota CAN sell their remainder to other polluters.
- One of the most common policy adopted by countries to meet their pledges under the Paris Agreement.

https://en.wikipedia.org/wiki/Carbon_emission_trading

Emissions Trading
EU-ETS

- EU’s Emissions Trading System (EU-ETS):
  - Currently covers ~40% of the EU’s emissions.
  - Includes power producers, aviation industry and manufacturing.
  - EU Commission proposed to include maritime emissions into the ETS in 2020.



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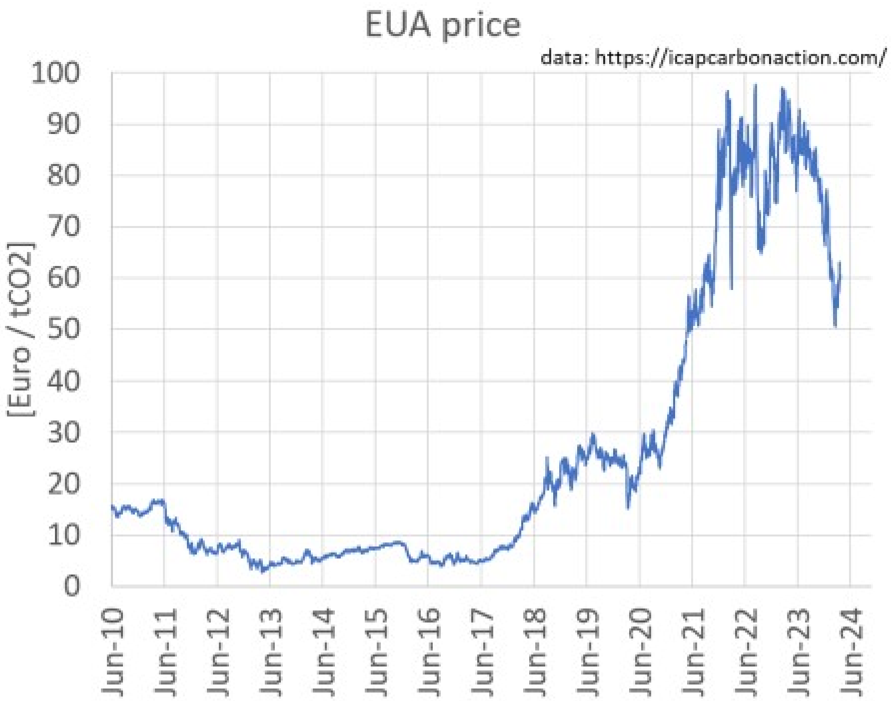

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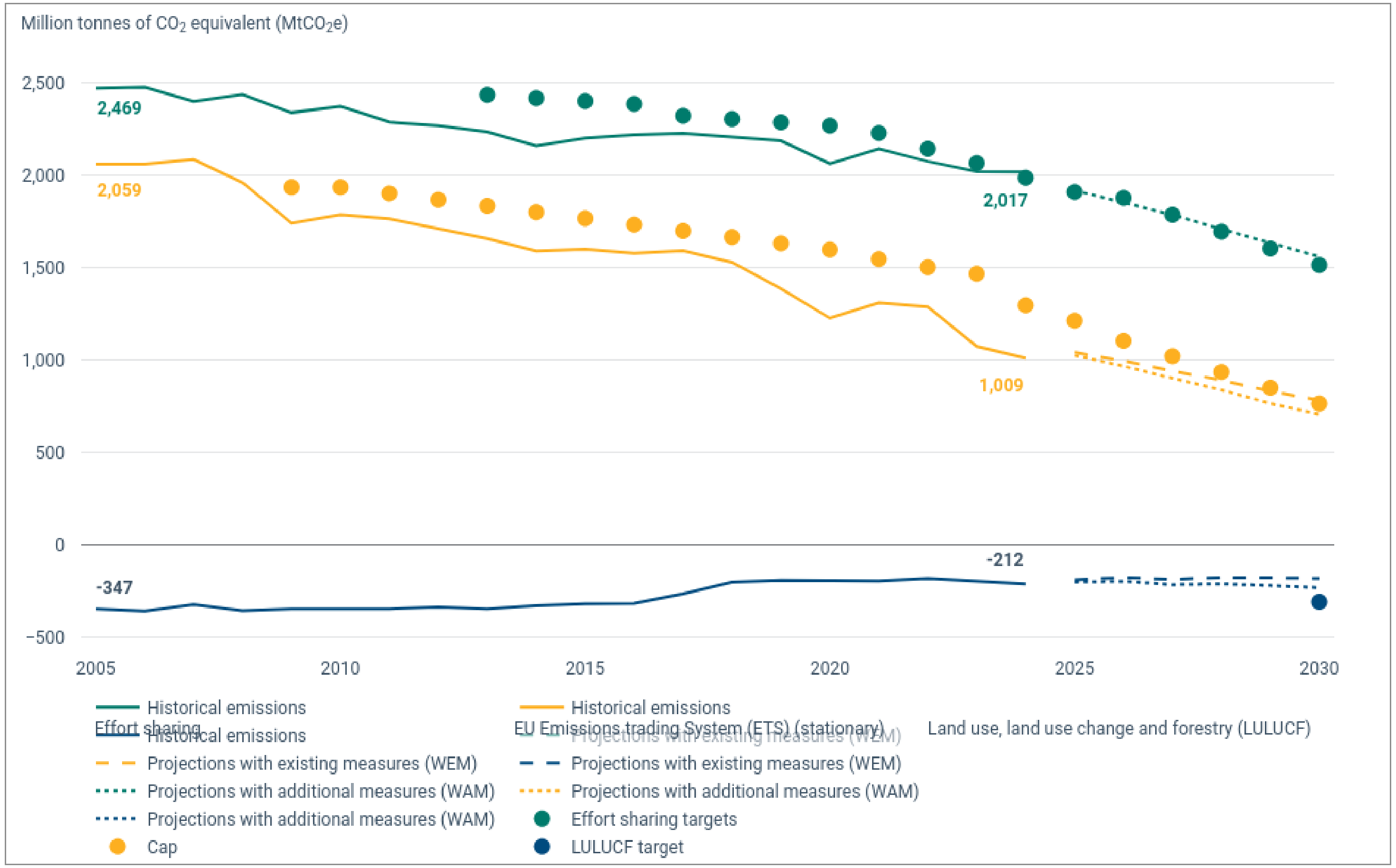
Emissions Trading

EU-ETS: Does it actually work?

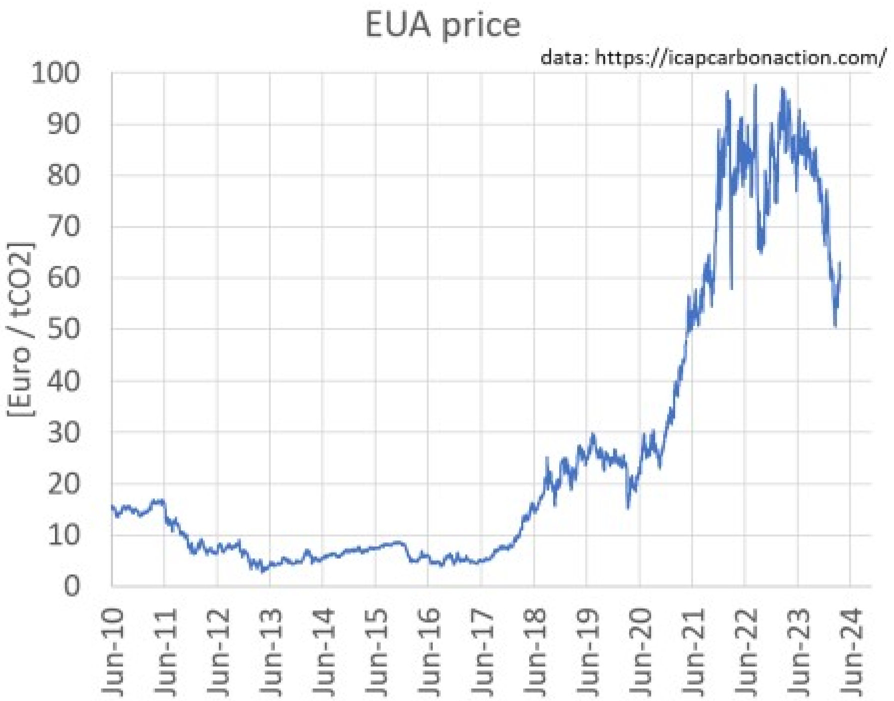


https://www.eea.europa.eu/en/analysis/indicators/total-greenhouse-gas-emission-trends

Emissions Trading

EU-ETS: Problems and Criticisms

- What about the remaining ~60%?
- At least 10 different types of fraudulent activities are possible:
  - Double counting
  - Exploitation of weak regulations
  - Tax fraud
- Oversupply of emissions allowances



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Mandaroux, Rahel, Chuanwen Dong, and Guodong Li. 2021. "A European Emissions Trading System Powered by Distributed Ledger Technology: An Evaluation Framework" Sustainability 13, no. 4: 2106. https://doi.org/10.3390/su13042106

Emissions Trading

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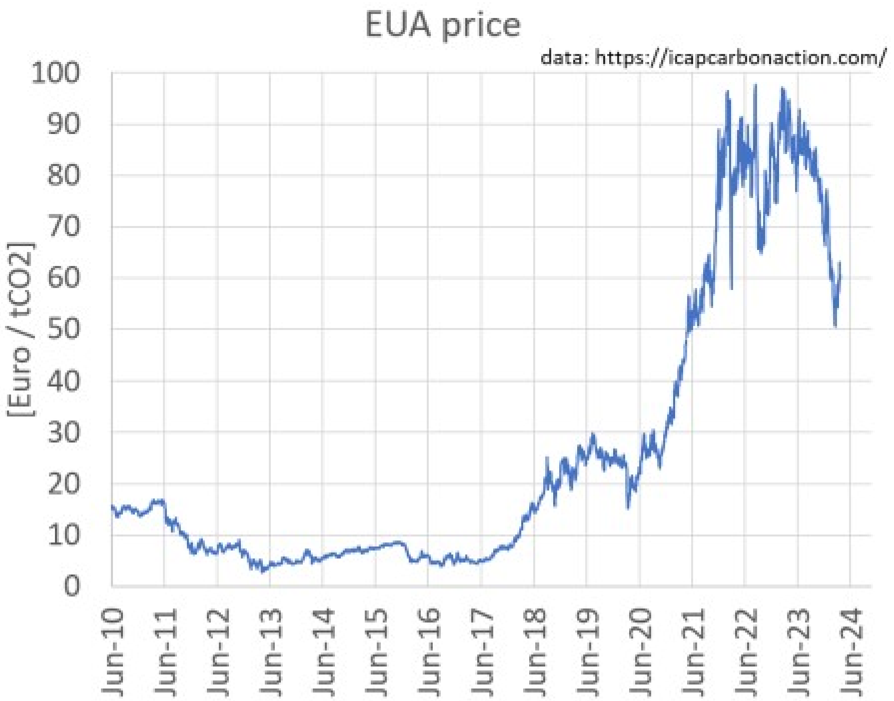

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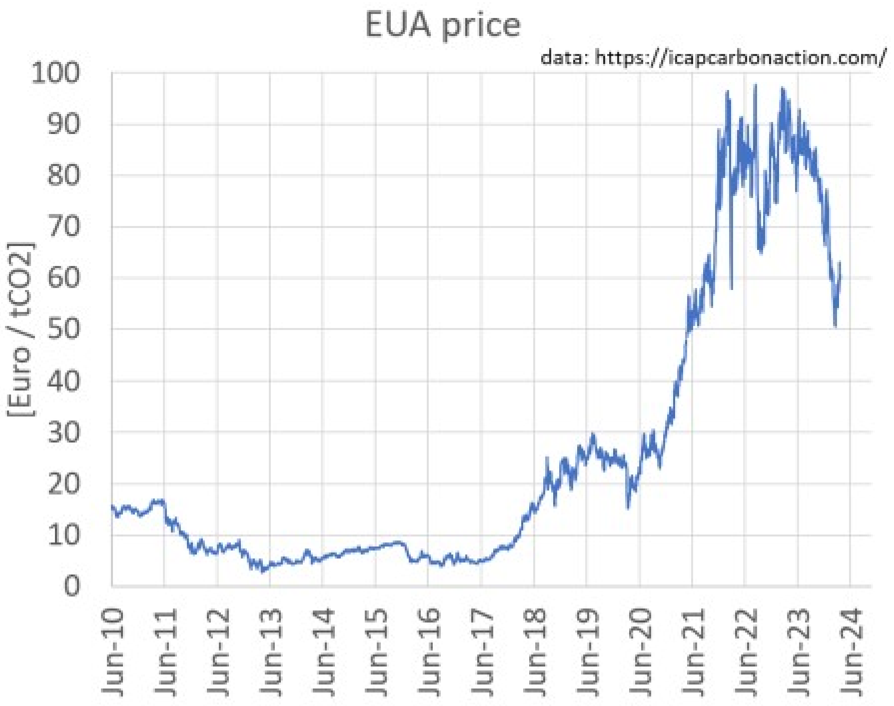
Emissions Trading

Quick Poll: What is the biggest source of emissions in the EU?



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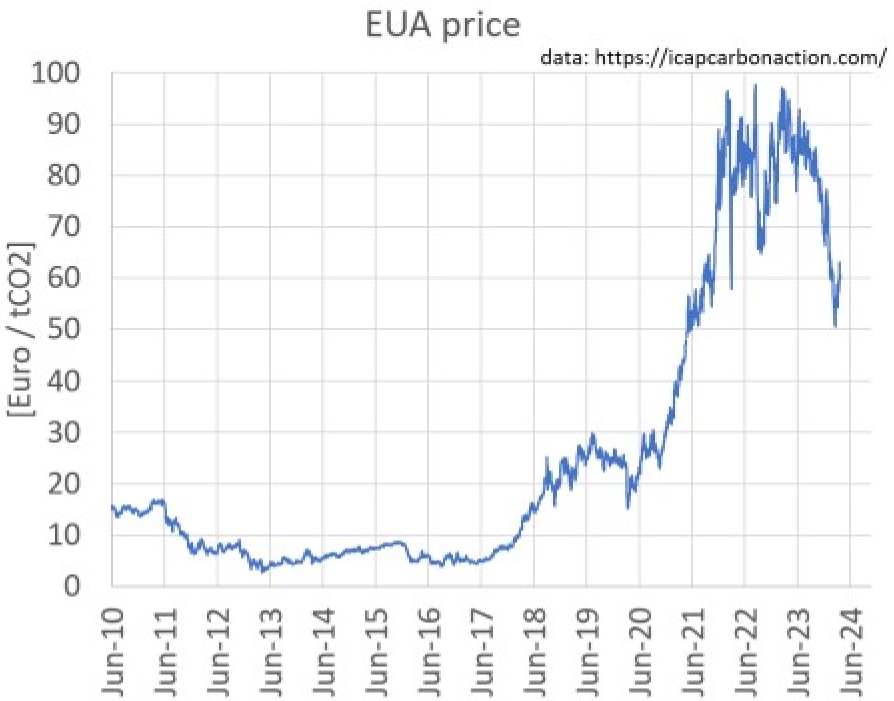
We Only Have One Planet
Environmental Pollution – Fossil Fuels

- 3 of the 10 dirtiest European coal plants are located in Poland
- In which country/countries are the other 7 dirtiest coal plants located?
  - 7 of the 10 dirtiest European coal plants are located in **GERMANY**

1. https://ember-climate.org/insights/research/top-10-emitters-in-the-eu-ets-2021/
2. John Englart – https://www.flickr.com/photos/takver/11308053925/ – CC BY-SA 2.0.
3. John Englart – https://www.flickr.com/photos/takver/51658831095/ – CC BY-SA 2.0.

Emissions Trading
EU-ETS

- EU's Emissions Trading System (EU-ETS):
  - Currently covers ~40% of the EU's emissions.
  - Includes power producers, aviation industry and manufacturing.



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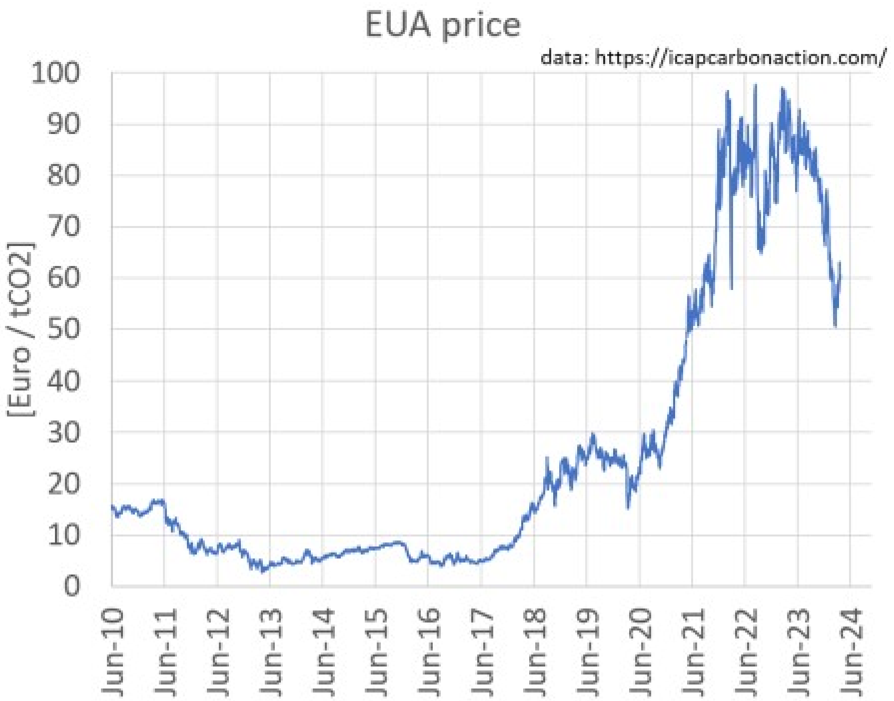
Emissions Trading

EU-ETS: Problems and Criticisms



- What about the remaining ~60% of emissions that are not covered by the EU ETS?
  - It's mostly energy
  - How is this possible?

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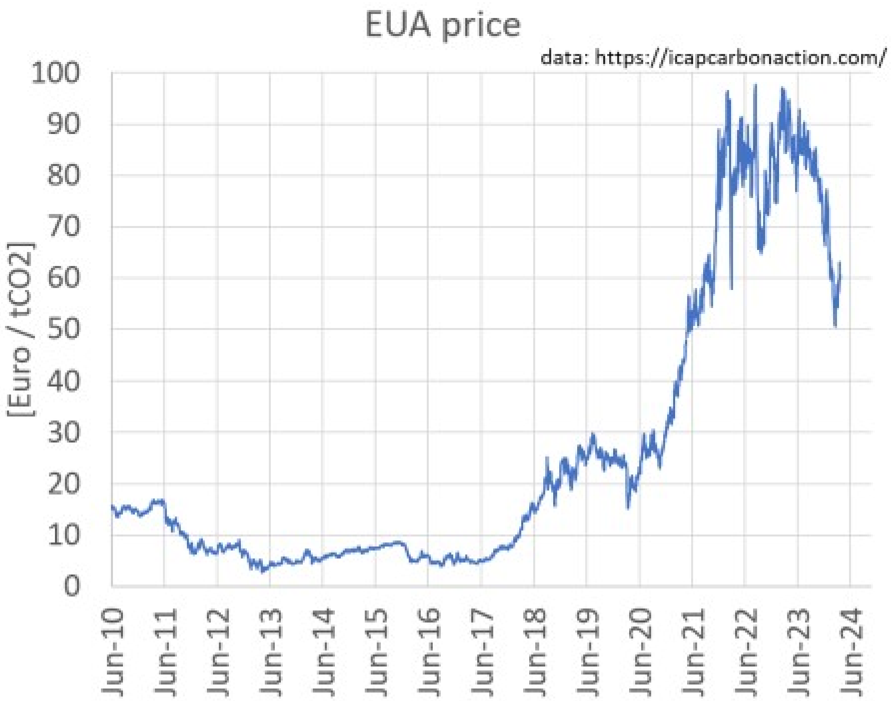
Emissions Trading
EU-ETS

- EU's Emissions Trading System (EU-ETS):
  - Currently covers ~40% of the EU's emissions.
  - Includes power producers, aviation industry and manufacturing ***(Within the Borders of the EU)***



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Emissions Trading

EU-ETS

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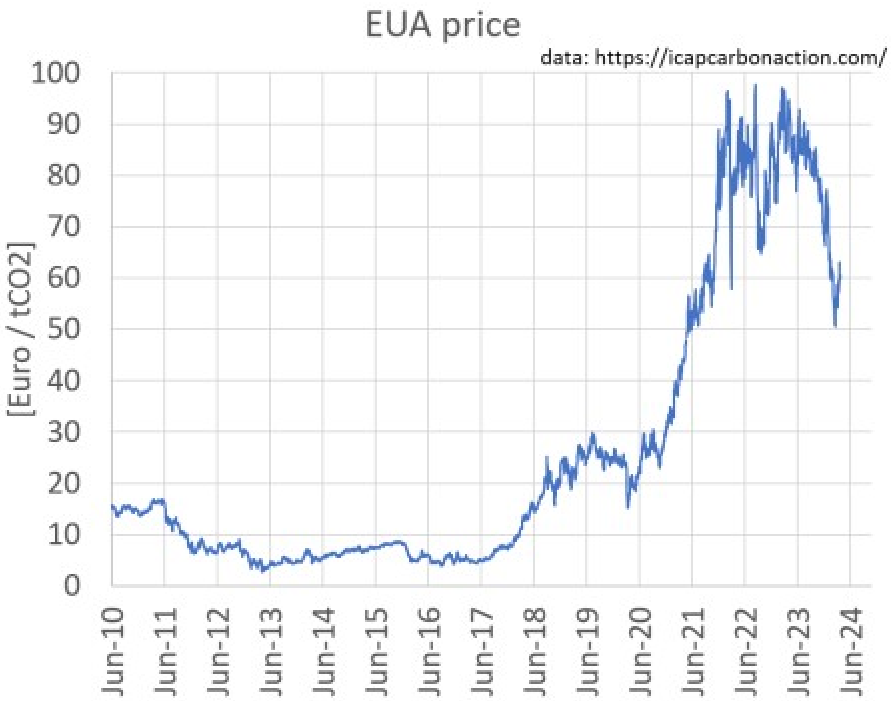
  - Includes power producers, aviation industry and manufacturing ***(Within the Borders of the EU)***
  - The EU imports most of it's oil and gas!



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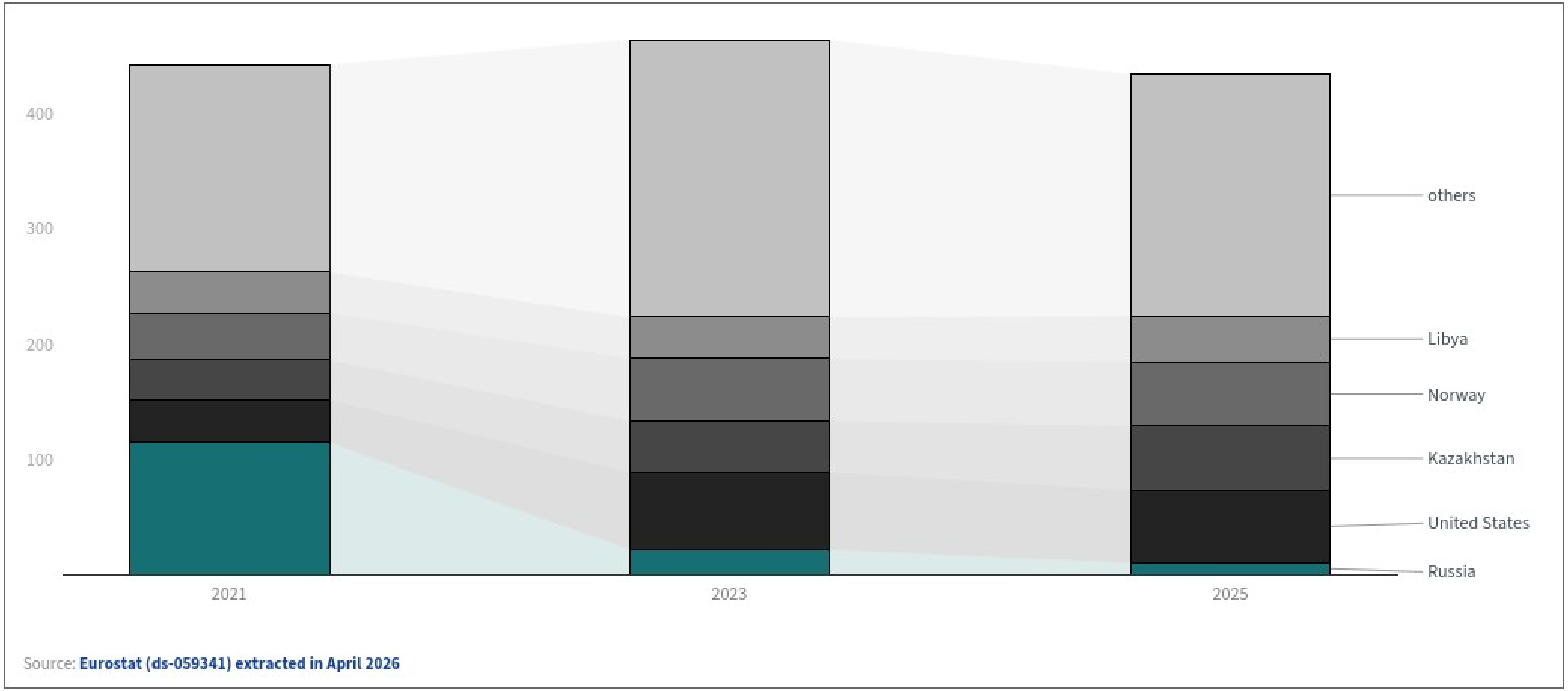

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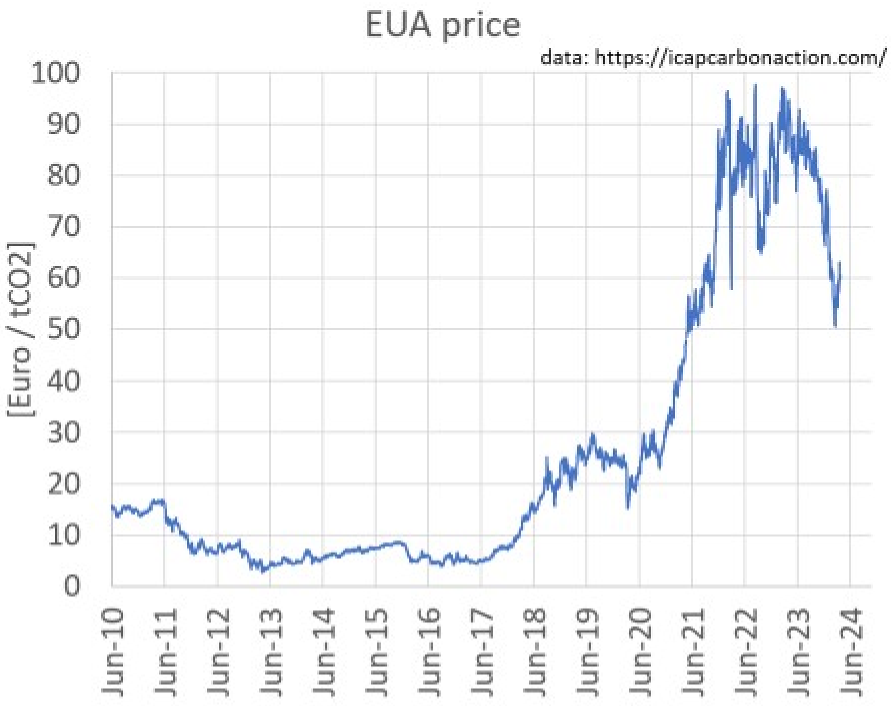
https://www.consilium.europa.eu/en/infographics/where-does-the-eu-get-its-oil-from/

Emissions Trading

EU-ETS: Problems and Criticisms

- When you import products from outside of the EU (a region with lower regulations), you don't always have to offset the emissions of manufacturing that product.



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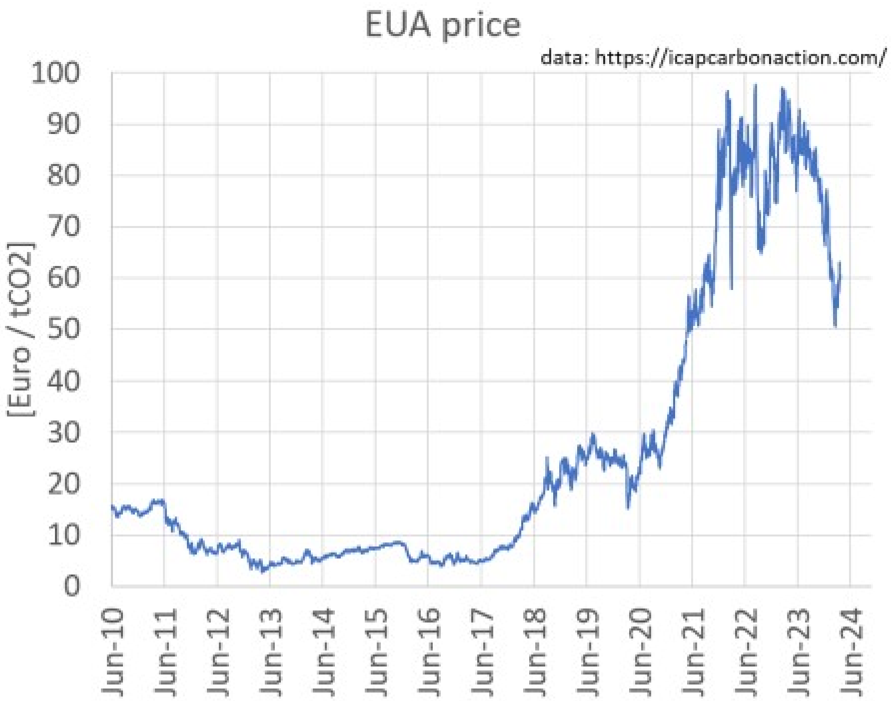
Emissions Trading

EU-ETS: Problems and Criticisms

- (Kind of) Good news: EU ETS-2 covers fossil fuel emissions by road transport (and other upstream emissions).
- Bad news: ETS-2 arrives in 2028.



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Two main perspectives

For any new technology *T*

Sustainability **of** *T*

VS

Usefulness of *T* **for** Sustainability

SUSTAINABILITY OF NEW EMERGING TECHNOLOGIES

Elephant(s) in the room: AI Datacenters

Gemini

aws

Claude

Elephant(s) in the room: AI Datacenters

- Microsoft said its carbon emissions had increased by 25% over the past year to 20m tonnes, “driven primarily by the expansion of our datacentre infrastructure”. (**up 56% since 2019**)
- Google said its emissions had increased 18% over the past year (to 18.8 million tonnes), “driven by increases in supply chain activities that supported the rapid expansion of our business”. (**up 100% since 2019**)
- Amazon reported a 16% increase in emissions overall (to ***80 million tonnes***), and a 20% increase in supply chain emissions, which included datacentre building and construction. In its report, it still framed this as “making progress” towards its goal of net zero emissions in 2040.

https://www.microsoft.com/en-us/corporate-responsibility/reports-hub

https://sustainability.google/reports/google-2026-environmental-report/

https://www.theguardian.com/us-news/2026/jul/11/microsoft-amazon-google-datacentre-carbon-emissions-france

Two main perspectives

Generative AI

Sustainability **of** *Generative AI*

VS

Usefulness of Generative AI **for** Sustainability

Two main perspectives

Generative AI

Sustainability **of** *Generative AI*

VS

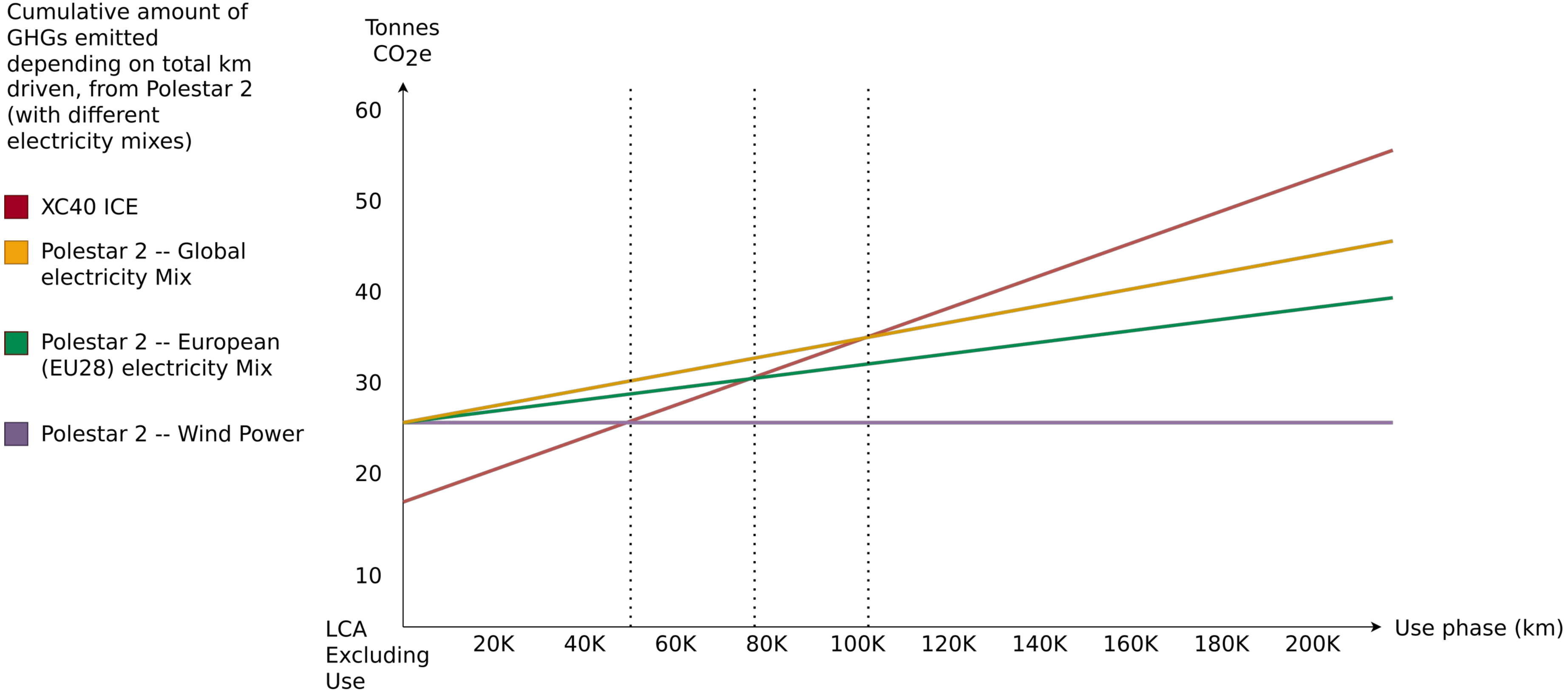
Usefulness of Generative AI **for** Sustainability

🤷

If you find out, let me know so I can put it on this slide.

Life Cycle Assessment – Polestar 2

Cumulative amount of GHGs emitted depending on total km driven, from Polestar 2 (with different electricity mixes)



Number of kilometers driven at break-even between Polestar 2 with different electricity mixes in the use phase of XC40 ICE (petrol)

Electric mix	Break-even (km)
Polestar 2 -- Global electricity Mix	112,000
Polestar 2 -- European (EU28) electricity Mix	78,000
Polestar 2 -- Wind Power	50,000

Polestar (2020) – Life Cycle Assessment – Carbon Footprint of Polestar 2.

Lifecycle Impact Assessment (LCIA)

2020 EU Study Example

Summary of overall lifecycle GWP impacts for Lower Medium Cars for different powertrain type

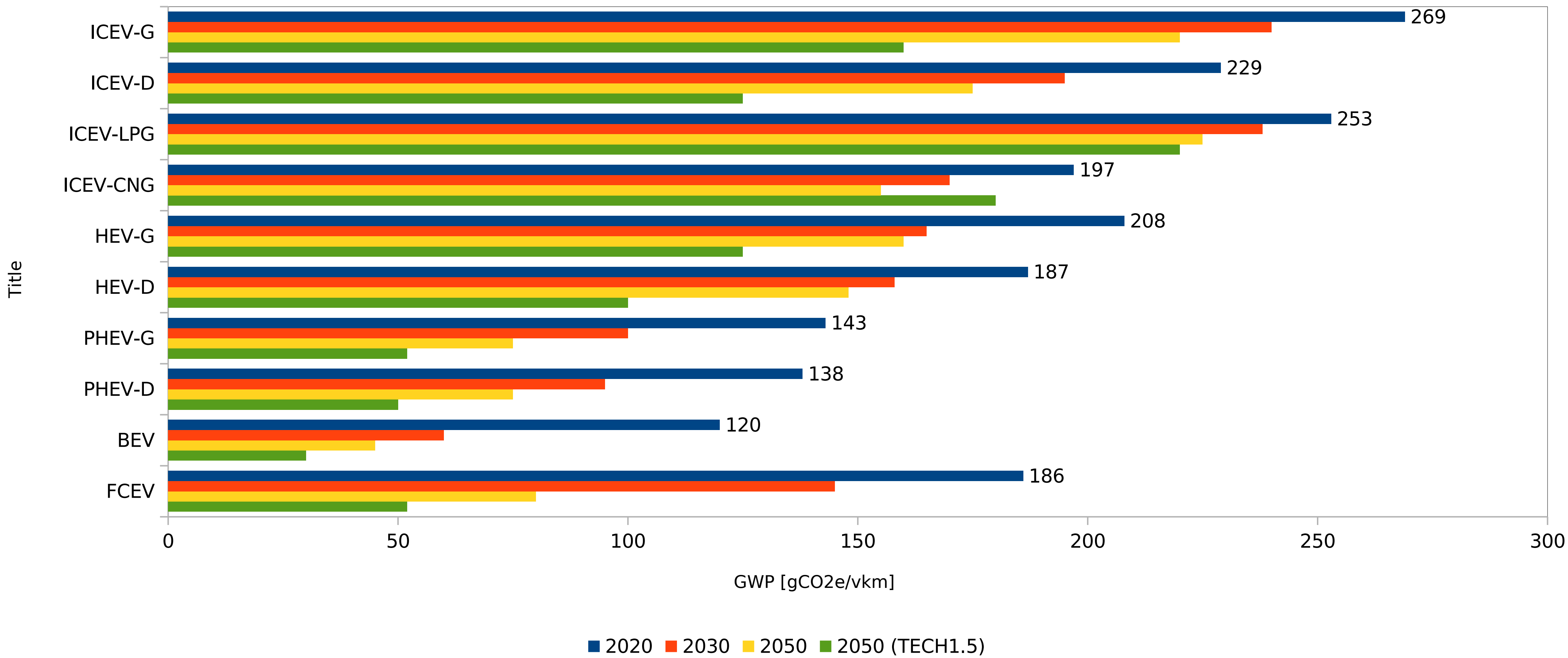


Chart adapted from Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA, Ricardo Energy and Environment (Link)

CE/CS and Technology

Does the tech-centric capitalist win?

	Holistic	Segmented
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Tire Microplastics: a problem that has worsened

- Microplastics is a problem, of which we have only seen the tip of the iceberg.
- A full understanding of the effects it has to the human body is not available yet, but what we do know, is quite bad.
- The biggest contributor to microplastic pollution is abrasion pollution from vehicle tires.

Vedula, Ravi T. "Tyre-Induced Microplastics: An Unknown Arena for e-Mobility." Premier Journal of Engineering 1 (2024): 100004.

Tire Microplastics: a problem that has worsened

- EV’s are significantly heavier than conventional vehicles, making tire abrasion almost twice as bad = almost twice as many microplastics emissions.
- People tend to accelerate more aggresively due to more available torque → more microplastic emissions.
- EVs constantly use regenerative braking instead of coasting like ICEs → more microplastic emissions.

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CE/CS and Technology

Can we do better than the tech-centric capitalist?

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CE/CS and Technology

Can we do better than the tech-centric capitalist?

	London	Paris	Strasbourg	Berlin	Rome
Aircraft	4.7	3.1	2.9	5.2	3.4
Train	1.0	1.0	1.0	1.0	1.0
EV	3.5	2.0	1.8	2.7	2.0
ICE	5	2.8	2.6	3.9	2.8

GHG emissions per passenger per trip relative to the emissions of a similar trip by train

Roosien RJ, Lim MNA, Petermeijer SM, Lammen WF. Multi-Modal Life Cycle Assessment of Journeys by Aircraft, Train or Passenger Car. Aerospace. 2024; 11(1):98.
https://doi.org/10.3390/aerospace11010098

CE/CS and Technology

Not All Change Is Progress

- Technologies change the world and our experience of living within it. But not all change is necessarily progress.
- Some changes may benefit one group while harming another, or benefit one goal at the expense of other goals.
- Such instances of change can only be considered true progress if we blind ourselves to these other negative effects.

https://consilienceproject.org/development-in-progress/

CE/CS and Technology
Progress Implies Better

- *Progress* is a statement about the world being in a different state. It is often the case that this difference in state is worse in some meaningful ways that may not be connected to our original intentions.

https://consilienceproject.org/development-in-progress/

CE/CS and Technology

Progress Implies Better

- *Progress* is a statement about the world being in a different state. It is often the case that this difference in state is worse in some meaningful ways that may not be connected to our original intentions.
  - (Maybe sometimes the original intentions might intentionally ignore the negative effects)

Gemini

aws

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https://consilienceproject.org/development-in-progress/

CE/CS and Technology
Progress Implies Better

- By defining progress too narrowly, we can call the positive outcomes in the here and now “progress,” while conveniently ignoring the harms occurring elsewhere.
- Minimizing negative externalities of technologies makes a safer, healthier, and ultimately better world for everyone alive now and for generations to come, who will inherit whatever we choose to leave them.

https://consilienceproject.org/development-in-progress/

CE/CS and Technology

How do we redefine progress?

For any new technology *T*

Environmental degradation **caused by** *T*

should not exceed

Usefulness of *T* **for** Environmental sustainability + human well-being

Conclusion

- Sustainability of a new technology vs do they help us become more sustainable?
- Innovation:
  - How do we get there? Reformist CS/ Techcentric CE / Transformational CS / Fortress CE
- Some examples on some “Emerging Technologies” but it might are not really change anything.
- Carbon Credits
- Generative AI
- EVs
- Additional theoretical notes:
  - https://consilienceproject.org/development-in-progress/

Questions?